MIL-C-43983(GL)
1 June 1978
SUPERSEDING
(See section 6)

MILITARY SPECIFICATION

CLOTH, KNITTED, POLYESTER, RIB KNIT

This specification is approved for use by Natick Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of polyester rib knit cloth.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AATCC TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 135 - 1973,
Dimensional Changes in Automatic Laundering of
Durable Press Woven or Knit Fabrics

(Application for copies should be addressed to AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709.)

3 REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for appearance, hand, shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. (see 6.4)

3.2.1 Fiber. The fiber shall be polyethelene glycol terephthalate, regular tenacity, semi-dull.

3.2.2 Yarn. The yarn shall be nominal 150 denier (17 tex) yarn which has been texturized by the false twist or friction twist method.

3.3 Knit. The knit construction of the cloth shall be as shown on figure 1.

3.4 Color. Unless otherwise specified, the color shall be Green 413 and shall be obtained by yarn dyeing.

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3.4.1 Matching. The color of the dyed and finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 kelvin.

3.4.2 Colorfastness. The dyed and finished cloth shall show fastness to laundering (after three cycles), crocking, perspiration, sublimation and light equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is available, the cloth shall show "good" fastness to laundering (after three cycles), crocking, perspiration, sublimation, and light.

3.5 Physical requirements. The physical requirements of the finished cloth shall be as shown in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Weight per sq. yd. min (g/m^2)	Texture per inch min (cm)	
Ounces	Wales	Courses
5.5 (187)	26 (10)	36 (14)

3.5.1 Width. Unless otherwise specified (see 6.2), the minimum width of the finished cloth shall be 60 inches (1524 mm).

3.6 Finish. The cloth shall be scoured, and heat set. The character of the finish shall be equal to that of the standard sample.

3.7 Dimensional stability. After 3 launderings, the finished cloth shall not shrink more than 3.0 percent in the length direction or 6.0 percent in the width direction, and shall not elongate more than 1 percent in either direction when tested as specified in 4.4.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2) the cloth shall be furnished in one continuous piece of not less than 50 yards (46 m) in length and shall be put-up in rolls, full width.

3.10 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

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3.11 Face identification. The face side of the cloth shall be identified by applying a stamping on that side with the word "Face" at each end of the roll.

3.12 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.5.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined on the face side and visual defects as defined in MIL-STD-1491 shall be classified as listed. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted as one defect for each yard (0.91 m) length-wise or fraction thereof in which it occurs. The sample unit shall be 1 yard (0.91 m).

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The lot size shall be expressed in units of 1 yard (0.91 m). The acceptable quality level shall be 4.0 defects per 100 units (yards). The sample size shall be in accordance with inspection level II of MIL-STD-105. The number of rolls from which the sample is to be selected shall be in accordance with table II. An approximate equal number of yards shall be examined in each roll in the sample.

Defects	
Abrasion mark	Misknit
Bowing	Missing yarn
Broken yarn	Mixed yarn
Cockled cloth	Needle line
Crack marks	Pin marks
Dirt	Permanent crease
Drop stitch	Run
Foreign matter	Skewing
Float	Snag
Heavy yarn	Spot or stain
Hole, cut, or tear	Thin yarn
Knot extending above surface of cloth	Tight yarn
Loose yarn	Tucking defect
Mend (visible)	Vertical lines
	Whiskering

4.2.2.2 Overall examination. Each defect listed shall be counted not more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size (number of rolls selected as sample) for this examination and the number of defects acceptable shall be as shown in table II. The rolls for this examination shall be those selected for yard-by-yard examination.

Defects

Character of finish not equal to that of the standard sample
Objectionable odor
Overall uncleanness
Uneven knitting throughout the roll
Edges frayed, ravelled, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll
Overall width less than specified
Appearance and hand not equal to that of the standard sample

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TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample
Up to 1300 1/	3	0
1301 up to and including 3200	5	0
3201 up to and including 8000	7	0
8001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	1
110,001 and over	25	1

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.8 m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.2.2.4 Examination for shade. Each roll in the lot shall be examined visually for shade (see 3.4.1). The sample unit shall be a piece not less than 5 inches by 8 inches (127 mm x 203 mm). A roll shall be unacceptable if the sample unit fails to meet the requirement for shade matching. A sample unit shall be drawn from each roll in the lot and identified with the roll number.

4.2.2.5 Examination for face marking and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have the face stamping missing from either or both ends or the face stamping on the wrong side, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of packaging. An examination shall be made in accordance with the provisions of PPP-P-1133, to determine that preservation-packaging, packing and marking comply with the section 5 requirements.

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4.4 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 continuous yards (2.7 m), full width of the finisned cloth. The lot size shall be expressed in units of 1 yard (0.91 m) each. The lot shall be unacceptable if one or more units fail to meet any test requirement specified the sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Identification of polyester	3.2.1	1/
Yarn denier	3.2.2	1/
Type of knit	3.3	1/
Colorfastness to:		
Laundering (after 3 cycles)	3.4.2	5610 4/
Crocking	3.4.2	5651
Perspiration	3.4.2	5680
Sublimation	3.4.2	5642 2/
Light	3.4.2	5660
Weight	3.5	5041
Wales and courses per inch (/cm)	3.5	5070
Finish		
Scoured	3.6	1/
Heat set	3.6	1/
Dimensional stability (after 3 cycles)	3.7	3/
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ The test shall be conducted at a temperature range of 5(205 ± 3°C).

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3/ AATCC Test Method 135 - 1973, Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabrics - Machine wash conditions Method II 120 ± 5°F (49 ± 3°C), drying procedures B (tumble).

4/ The specimen must be dried after each of the 3 laundering cycles. On the color transfer cloth evaluation, only the stain of the polyester segment shall be considered.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C packing. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The knitted cloth is intended for use in the manufacture of women's uniforms.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth required if other than specified (see 3.4).
(c) Width required when other than specified (see 3.5.1).
(d) Length required if other than specified (see 3.9).
(e) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

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6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Supersession data. This specification includes the requirements of Purchase Description LP/P. DFS 8-76, Cloth, Knitted, Polyester, Tab Knit, dated 4 May 1976.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

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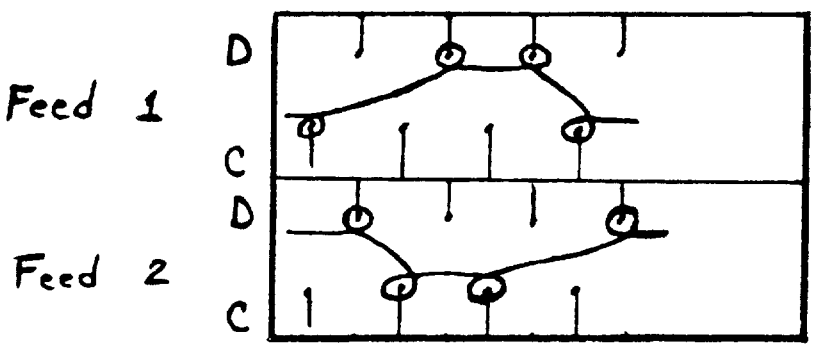


Figure 1 - Knit Construction

Key D — Dial Needles
C — Cylinder Needles

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DOCUMENT IDENTIFIER AND TITLE MIL-C-43983(GL)
CLOTH, KNITTED, POLYESTER, REP KNIT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
YES NO (If "Yes" in what way?)

4 REMARKS

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